

Lipid-Mediated Delivery of Oligonucleotide to Pulmonary Endothelium

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Pulmonary endothelium plays an important role in the maintenance of normal pulmonary physiology and its dysfunction is involved in a number of pulmonary diseases. Correction of endothelial dysfunction via antisense oligodeoxynucleotides (ODN) is dependent on the development of a delivery vehicle that can efficiently deliver the ODN to pulmonary endothelium with minimal toxicity. To this end, we have developed a lipidic vector (LPD) that is composed of 1,2-dioleoyl-3-trimethylammonium-propane (DOTAP) liposomes, protamine, and ODN. This formulation is highly efficient in delivering ODN to the lung via the vascular route. The efficiency of delivery is a function of lipid composition and of the charge ratio between lipid and ODN. Immunofluorescence staining of BrdU-labeled ODN suggested efficient accumulation of ODN in the alveolar capillary region. Transmission electron microscopy of immunogold localization of BrdU-labeled ODN confirmed that pulmonary endothelial cells were indeed targeted by the vector. Furthermore, this formulation is associated with minimal proinflammatory cytokine response and other hematologic toxicities when the ODN lack a potent unmethylated CpG motif. Pretreatment of mice with LPD containing an ODN against intercellular adhesion molecule-1 (ICAM-1) significantly decreased ICAM-1 expression in the lung following LPS challenge. These results provide a basis for lipid-mediated delivery of ODN for the treatment of pulmonary diseases.

A number of pulmonary diseases, including acute respiratory distress syndrome, involve endothelial dysfunction that is associated with an increased expression of several genes (1, 2). Downregulation of the "disease" genes thus represents an attractive approach for the treatment of these pulmonary disorders. Antisense oligodeoxynucleotides (ODN) have attracted increasing attention as a therapeutic agent because they can specifically downregulate the expression of a gene by forming a heteroduplex with its mRNA (3).

Antisense technology also holds potential for the study of gene function related to lung physiology (4). ODN can be readily synthesized in large quantities and thus can be used to characterize a large number of sequences emerging from genome projects in a cost-effective manner. In addition, organ-specific downregulation of a gene can be achieved via the use of ODN in combination with a tissue-

specific delivery vehicle. Finally, the degree and duration of gene suppression can be modulated by adjusting the dose and frequency of ODN administration. Recently, several new types of ODN have been developed, such as peptide nucleic acids (5) and mixed backbone oligonucleotides (6, 7) that demonstrate improved antisense activity and reduced *in vivo* toxicity.

The success of these studies, however, is largely dependent on, among other things, the development of a delivery vehicle that can efficiently deliver the ODN to pulmonary endothelium with minimal toxicity. To this end, we have developed a novel lipidic vector composed of DOTAP liposomes, protamine, and ODN that is highly efficient in delivering ODN to the lung via the vascular route. Immunohistochemical studies show that endothelial cells are the major cell type that takes up the ODN. Furthermore, we have shown that this formulation is associated with minimal proinflammatory cytokine response and other hematologic toxicities when the ODN lack a potent unmethylated CpG motif(s). These results provide a basis for lipid-mediated delivery of ODN for the treatment of pulmonary diseases. They also suggest the utility of this approach as a research tool to characterize the function of novel genes in pulmonary endothelium.

Materials and Methods

Chemicals

DOTAP and dioleoylphosphatidyl-ethanolamine (DOPE) were purchased from Avanti Lipids Inc. (Alabaster, AL). Cholesterol was obtained from Sigma (St. Louis, MO). Protamine sulfate-USP was from Eli Lilly (Indianapolis, IN). Na¹²⁵I was from Dupont NEN (Boston, MA). All other chemicals were of reagent grade.

ODN Synthesis

Antisense ODN were purchased from MWG Biotech Inc. (High Point, NC) and their sequences are shown in Table 1. BrdU-labeled ODN were obtained from Midland Certified Reagent Co. (Midland, TX).

Iodination of ODN

ODN were iodinated as described (8). Briefly, ¹²⁵I was added to ODN in 0.1 mM sodium acetate (pH 4.0) in a glass tube coated on the bottom with Iodogen reagent (Pierce, Rockford, IL). The mixture was incubated in a water bath at 50°C for 30 min. Iodinated ODN were then purified from free iodine by passing through a Bio-Gel P-4 spin column (Bio-Rad, Richmond, CA).

Preparation of Liposomes

Liposomes containing DOTAP alone or in a 1:1 molar ratio with cholesterol or DOPE were prepared as follows. The lipid mixture in chloroform was dried as a thin layer in a 100-ml round-bottomed flask which was further dried under vacuum for 2 h. The

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Abbreviations: dioleoylphosphatidyl-ethanolamine, DOPE; 1,2-dioleoyl-3-trimethylammonium-propane, DOTAP; intercellular adhesion molecule-1, ICAM-1; interferon, IFN; interleukin, IL; lipidic vector, LPD; oligodeoxynucleotides, ODN; PBS containing 0.5% bovine serum albumin and 0.15% glycine, PBG; phosphate-buffered saline, PBS.

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TABLE 1
Sequences of ODN used

ODN	Sequence
Endo-AS-1	5' ATA ATC CAT TCT GCA AAG GG 3'
Endo-AS-2	5' AAT AAT CCA TTC TGC AAA GG 3'
Endo-AS-3	5' AAA TAA TCC ATT CTG CAA AG 3'
Endo-AS-4	5' AAA ATA ATC CAT TCT GCA AA 3'
Endo-AS-5	5' GAA AAT AAT CCA TTC TGC AA 3'
Endo-AS-6	5' GGA AAA TAA TCC ATT CTG CA 3'
Endo-AS-7	5' GGG AAA ATA ATC CAT TCT GC 3'
Endo-AS-8	5' CGG GAA AAT AAT CCA TTC TG 3'
Endo-AS-9	5' ACG GGA AAA TAA TCC ATT CT 3'
Endo-AS-10	5' CAC GGG AAA ATA ATC CAT TC 3'
1668-PS	5' TCC ATG ACG TTC CTG ATG CT 3'
ISIS 3082	5' TGC ATC CCC CAG GCC ACC AT 3'
ISIS 1082	5' GCC GAG GTC CAT GTC GTA CGC 3'

lipid film was hydrated in 5% dextrose in water to give a final concentration of 10 mg DOTAP/ml. Preparation of small unilamellar vesicles by extrusion was performed as follows. The lipid solution was briefly sonicated, followed by incubation at 50°C for 10 min, and then sequentially extruded through polycarbonate membranes with the following pore sizes: 1.0, 0.6, and 0.2 μm . The size of liposomes was measured by dynamic laser scattering using a Coulter N4SD particle sizer (Hialeah, FL).

Preparation of Lipidic Vector

For preparation of lipidic vector (LPD), various amounts of liposomes and protamine sulfate were mixed in 5% dextrose. ODN, diluted with 5% dextrose, were then added to liposome/protamine solution. The mixture was allowed to stand at room temperature for 10 min before use. LPD was injected into mice through the tail vein (200 μl /mouse).

Effect of Mouse Serum on the Release of ODN from LPD

LPDs of different lipid compositions were mixed with mouse serum (Charles River Laboratories, Wilmington, MA) at a 1:2 (vol/vol) ratio and the mixtures were incubated at 37°C with gentle shaking. Aliquots of samples were collected at 15, 45, and 75 min following the incubation and then subjected to centrifugation at 14,000 rpm for 10 min. The ODN that were released into the serum were determined by electrophoresis on an agarose gel.

In Vivo Distribution of LPD

Female CD-1 mice, 4–6 wk of age, were purchased from Charles River Laboratories and housed in accordance with institutional guidelines. ^{125}I -labeled LPDs of different lipid compositions were administered into mice at a dose of 25 μg ODN per mouse via tail vein in 200 μl of 5% dextrose. At different times following injection, mice were killed. Blood and major organs were collected, assayed for radioactivity, and the result was expressed as the percentage of injected dose per organ.

In a separate experiment, LPD containing BrdU-labeled ODN was injected intravenously into mice at a dose of 25 μg ODN per mouse. At 30 min after injection, mice were killed and lungs were perfused intravascularly with phosphate-buffered saline (PBS) followed by 2% paraformaldehyde in PBS, and inflated with this fixative to near total lung capacity. The lungs were rinsed with cold PBS and immersed in 30% sucrose in PBS at 4°C overnight. The lungs were then quickly frozen in OCT with dry ice. Five-micrometer lung cryosections were then cut, permeabilized with Triton X-100 (0.2% vol/vol in PBS) for 10 min at room temperature (RT), and washed three times with PBS. Following three washes in PBS containing 0.5% bovine serum albumin and 0.15%

glycine (PBG buffer) sections were incubated in a 1:100 dilution of biotin-conjugated mouse anti-BrdU (Zymed Laboratories Inc., South San Francisco, CA) for 1 h at RT, washed with PBG three times, and labeled with Streptavidin Alexa 488 (Molecular Probes, Eugene, OR) for 1 h at RT. Following three further washes with PBG the sections were stained with Rhodamine Phalloidin for 40 min (Molecular Probes) to label the actin cytoskeleton. Sections were washed with PBG three times and stained with Hoescht dye 33,258 (Sigma) for 30 s and mounted in Gelvatol (Monsanto, St. Louis, MO). Cells were visualized using an Olympus Provis microscope (Olympus, Tokyo, Japan) using a triple pass (blue/green/red) cube, which allows excitation at 384 nm and collection at 540 nm. Images were collected using an Optronics Magnifier Camera (Santa Barbara, CA) or with a Leica TCS NT confocal microscope with a $\times 60$ oil immersion objective at a $1,024 \times 1,024$ pixel resolution.

Immunoelectron Microscopy

LPD containing BrdU-labeled ODN was injected intravenously into mice at a dose of 25 μg ODN per mouse. At 5 and 30 min after injection, mice were killed and lungs were fixed with 2% paraformaldehyde, 0.01% glutaraldehyde in 0.1 M PBS, and stored at 4°C for 1 h. Pieces (1-mm³ in size) of the fixed lung tissue were infused with 2.3 M sucrose in 0.1 M PBS overnight at 4°C. Tissue was frozen on ultracytome stubs under liquid nitrogen and stored in liquid nitrogen until use. Ultrathin sections (70–100 nm) were cut using a Reichert Ultracut U ultramicrotome with a FC4S cryo-attachment (VELP Scientific Inc., Valencia, CA), lifted on a small drop of 2.3 M sucrose, and mounted on Formvar-coated copper grids. Sections were washed three times with PBS, then three times with PBG. Sections were labeled with a 1:100 dilution of biotin-conjugated mouse anti-BrdU (Zymed Laboratories Inc.) in PBG buffer for 1 h at RT. Sections were washed 4 $\times$ for 5 min in PBG buffer, then labeled with a 1:25 dilution of streptavidin conjugated to 5 nm colloidal gold (Amersham, Piscataway, NJ) (1:25) at RT for 1 h. Sections were washed three times in PBG, three times in PBS, then fixed in 2.5% glutaraldehyde in PBS for 5 min, washed two times in PBS, then washed six times in ddH₂O. Sections were post-stained in 2% neutral uranyl acetate for 7 min, washed three times in ddH₂O, stained for 2 min in 4% uranyl acetate, then embedded in 1.25% methyl cellulose. Labeling was observed on a JEOL JEM 1,210 electron microscope (Peabody, MA) at 80 kV.

In Vivo Toxicity Assays

Groups of six mice received tail vein injection of dextrose, DOTAP liposome/protamine, or LPD containing either ODN 8 or ODN 1,668. The doses for DOTAP, protamine, and ODN were 212 μg , 7.5 μg , and 25 μg per mouse, respectively. Twenty-four hours after injection, mice were anesthetized and blood was collected by retro-orbital bleeding. Blood was collected into microtainers containing EDTA to determine the complete blood count (CBC), which included a white blood cell count with differential, a red blood cell count, hemoglobin, and hematocrit. Blood was also collected into serum separator tubes for a serum chemistry profile. The serum chemistry profile included determinations of serum transaminases (alanine aminotransferase [ALT] and aspartate aminotransferase [AST]), alkaline phosphatase, creatinine kinase, bilirubin, serum protein levels including albumin and globulin levels, blood urea nitrogen, amylase, glucose, and electrolytes. Whole blood and serum samples were analyzed by the Antech Diagnostics (Farmingdale, NY).

In a separate experiment, groups of six mice received tail vein injection of LPD containing ODN of different sequences at a dose of 25 μg /mouse. Mice were bled from retro-orbital sinuses under anesthesia 2 h after the injection. Serum levels of mouse

tumor necrosis factor (TNF)- α were determined with a specific immunoassay kit for mouse TNF- α (R&D, Minneapolis, MN).

The Effect of LPD on Lung Permeability

The effect of LPD on lung permeability was examined by determining the lung wet/dry weight ratio 12 h after intravenous administration of LPD (9). Groups of six mice received tail vein injection of DOTAP liposome, or LPD containing either ODN 8 or ODN 1,668. The doses for DOTAP, protamine, and ODN were 212 μ g, 7.5 μ g, and 25 μ g per mouse, respectively. Twelve hours after injection, mice were bled from retro-orbital sinuses and then killed by cervical dislocation. Lungs were excised *en bloc* and dissected away from the heart and thymus. The lungs were immediately weighed and then placed in a desiccating oven at 65°C for 48 h, at which point dry weight was achieved. The ratio of wet/dry weight (g water/g lung dry weight) was used to quantify lung water content.

The Effect of Intercellular Adhesion Molecule-1 Antisense ODN on *Escherichia coli* Endotoxin-Induced Expression of Intercellular Adhesion Molecule-1 mRNA

CD-1 mice received an intravenous injection of intercellular adhesion molecule (ICAM)-1 antisense ODN (ISIS 3,082) or control ODN (ISIS 1,082) formulated in LPD at a dose of 2.5 mg/kg. Thirty minutes after administering the ODN, mice were anesthetized via intraperitoneal injection of 2,2,2-tribromoethanol (400 mg/kg), and endotoxin (80 μ l of 2.5 mg/ml) was instilled into the airways through the trachea. Control mice received saline. Four hours after the instillation, mice were killed and lungs were collected. Total RNA was isolated from lungs using RNAqueous kit (Ambion Inc., Austin, TX). Total RNA was separated on agarose/formaldehyde gels and transferred to nylon membranes. Northern blots were probed for ICAM-1 and β -actin mRNA using random primed labeled cDNA probes. RNA was quantitated using a Personal Molecular Imager FX (Bio-Rad, Hercules, CA). The increase in ICAM-1 was normalized for differences in loading using the β -actin mRNA.

Statistical Analysis

Data were expressed as means $\pm$ standard deviation and analyzed by the two-tailed unpaired Student *t* test using the PRISM software program (GraphPad Software, San Diego, CA). Data were considered significant if $P < 0.05$ (*) and very significant if $P < 0.01$ (**).

Results

Pulmonary Uptake of ODN is Lipid Dose-Dependent

Figure 1 shows the pulmonary uptake of ODN with increasing amounts of DOTAP. The lungs were collected at 30 min following tail vein injection of LPD. Increasing the amount of DOTAP from 2.1–8.5 μ g lipid/ μ g ODN led to a significant increase in ODN uptake. Further increase in DOTAP was associated with a much smaller increase in ODN uptake. Thus, the dose of 8.5 μ g DOTAP/ μ g ODN was used in subsequent studies.

Pulmonary Uptake of ODN is Lipid Composition-Dependent

Previous studies have shown that the lipid composition greatly affects the efficiency of cationic lipid-mediated gene transfer to the lung via the vascular route (10–12). Figure 2 shows that the efficiency of ODN delivery to the lung via LPD is also significantly affected by the lipid composition. Inclusion of a neutral lipid, particularly DOPE, into DOTAP liposomes

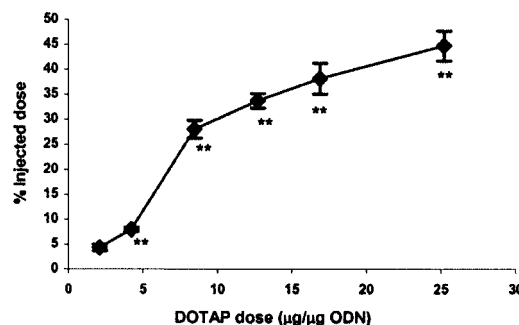


Figure 1. Pulmonary uptake of ODN as a function of the dose of DOTAP lipid. LPD was prepared by mixing of protamine (0.3 μ g protamine/ μ g ODN) and various amounts of DOTAP liposomes followed by the addition of 125 I-labeled ODN. LPD was then injected into mice through tail vein at a dose of 25 μ g ODN/mouse. Thirty minutes after the injection, the mice were bled from retro-orbital sinuses under anesthesia and then killed by cervical dislocation. Lungs were collected, weighed, and assayed for radioactivity and the results were expressed as the percentage of injected dose per organ. ** $P < 0.01$ (versus 2.1 μ g lipid/ μ g ODN). $n = 6$ animals.

led to a significant decrease in ODN uptake. LPD with no neutral lipid is the most efficient vector for ODN delivery. This is different from DNA delivery, where cholesterol-containing LPD is the most efficient vector (10–12).

As an approach to understand why LPDs of different lipid compositions vary in the efficiency of ODN delivery, we studied their sensitivity to mouse serum with respect to ODN release. Significantly greater amounts of free ODN were detected in the DOPE formulation compared with LPDs containing cholesterol or no neutral lipid (Figure 3). The amount of free ODN released from LPD increased with incubation time. LPD containing no neutral lipid appeared to be the most stable formulation; no free ODN were detected even 75 min after exposure to serum (Figure 3). The excellent stability might explain the efficient ODN delivery to the lung by this formulation. Thus, the latter formulation was used in subsequent distribution and toxicity studies.

Tissue Distribution of ODN following Intravenous Injection of LPD

Figure 4 shows the distribution of ODN in several major organs at different times following tail vein injection of LPD. LPD with no neutral lipid was used and ODN were labeled with 125 I. Initially, ODN were mainly localized in the lung. Over time the ODN were gradually removed from the lung. The low recovery of ODN in major organs at later time points was probably due to a rapid degradation of ODN in the liver. Little ODN were found in heart, spleen, and kidney. Figure 5 shows the distribution of BrdU-labeled ODN in lung sections at 30 min following LPD injection. ODN were mainly accumulated in alveolar capillary region (B and C). Confocal analysis of lung sections suggested that majority of the BrdU-ODN observed were inside cells (D). ODN were also observed inside the endothelial cells that are lining the small vessels (D). No false positive signals were found in lung sections of control mice that received nonla-

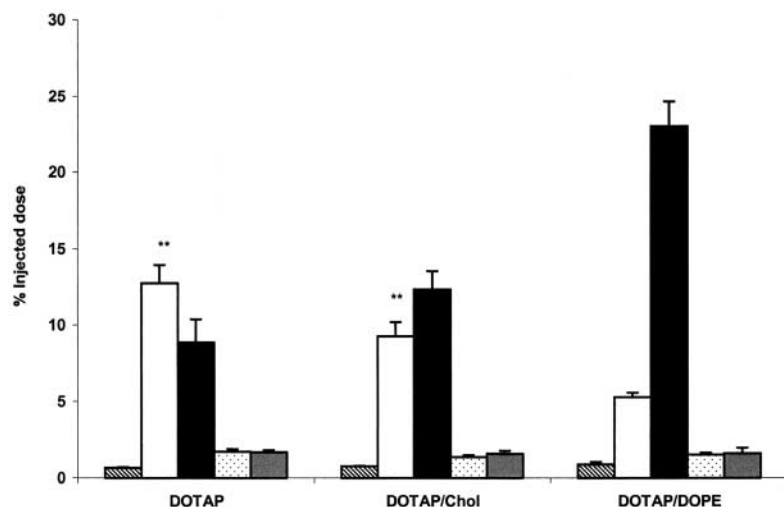


Figure 2. Effects of lipid composition on pulmonary uptake of ODN. Liposomes composed of DOTAP:cholesterol (m/m, 1/1), DOTAP:DOPE (m/m, 1/1) or DOTAP alone were prepared by an extrusion method. They were then used to prepare LPD, respectively, as described in the legend to Figure 1, and 25 μ g of ODN was injected intravenously into each mouse. One hour after injection, mice were killed and ODN distribution in major organs was analyzed. *Striped bars*, heart; *open bars*, lung; *filled bars*, liver; *dotted bars*, spleen; *shaded bars*, kidney. ** $P < 0.01$ (versus DOTAP/DOPE). $n = 6$ animals.

beled ODN (A). This experiment was repeated three times and similar results were obtained in these studies.

Immunoelectron Microscopic Examination

Uptake of ODN by pulmonary endothelium was confirmed by immunoelectron microscopic examination of lung sections at 5 and 30 min following the injection of LPD containing BrdU-labeled ODN. BrdU-labeled ODN were visualized by staining with biotin-conjugated mouse anti-BrdU IgG followed by streptavidin-conjugated to 5 nm colloidal gold. Five minutes after the injection of LPD, ODN were localized largely intravascularly and associated with the lipidic vector (Figures 6A and 6B). These ODN, however, were tightly associated with capillary wall as they could

not be washed away by extensive intravascular perfusion. Thirty minutes after the injection, some ODN were found to be in close contact with endothelial cells and invagination of ODN-containing LPD by endothelial cells was visualized (Figures 6C and 6D). ODN were clearly observed in the cytosol of endothelial cells (Figures 6C–6F). ODN were also observed in other types of cells such as macrophages but at much lower frequencies. No gold particles were seen in lung sections of mice that received nonlabeled ODN (data not shown). This experiment was repeated three times and similar results were obtained in these studies.

ODN without CpG Motifs Are Not Proinflammatory following Systemic Delivery by LPD

One of the problems that is associated with cationic lipid-mediated delivery of nucleic acids (DNA and ODN) is the induction of proinflammatory cytokines such as IL-1, IL-12, IFN- γ , and TNF- α (13, 14). Cytokine induction is largely

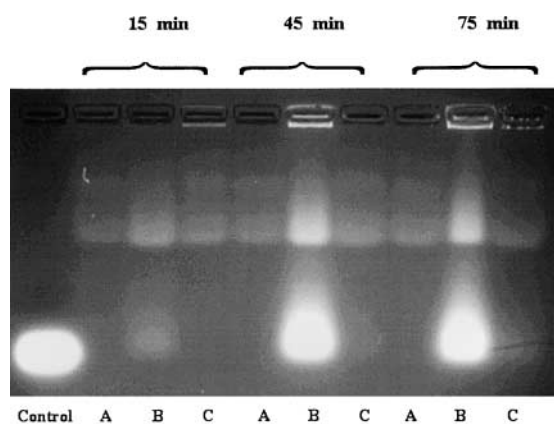


Figure 3. Release of ODN after exposure of LPDs to mouse serum. LPDs of different lipid compositions were mixed with serum as described in MATERIALS AND METHODS. Aliquots of samples were collected at 15, 45, and 75 min following the incubation and then subjected to centrifugation at 14,000 rpm for 10 min. The ODN that were released into the serum were determined by electrophoresis on an agarose gel. (A) LPD containing DOTAP alone; (B) LPD containing DOTAP:DOPE; (C) LPD containing DOTAP:cholesterol.

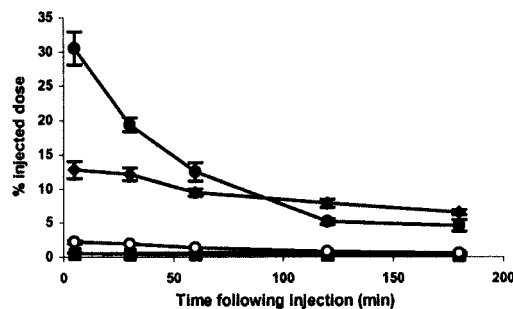


Figure 4. *In vivo* distribution of ODN at different times following intravenous injection. 125 I-labeled ODN were formulated in LPD composed of DOTAP liposomes and injected into mice at a dose of 25 μ g ODN/mouse. At different times following the injection, mice were killed and the uptake of ODN by major organs was analyzed as described in the legend to Figure 1. *Filled circles*, lung; *diamonds*, liver; *squares*, heart; *triangles*, spleen; *open circles*, kidney. $n = 6$.

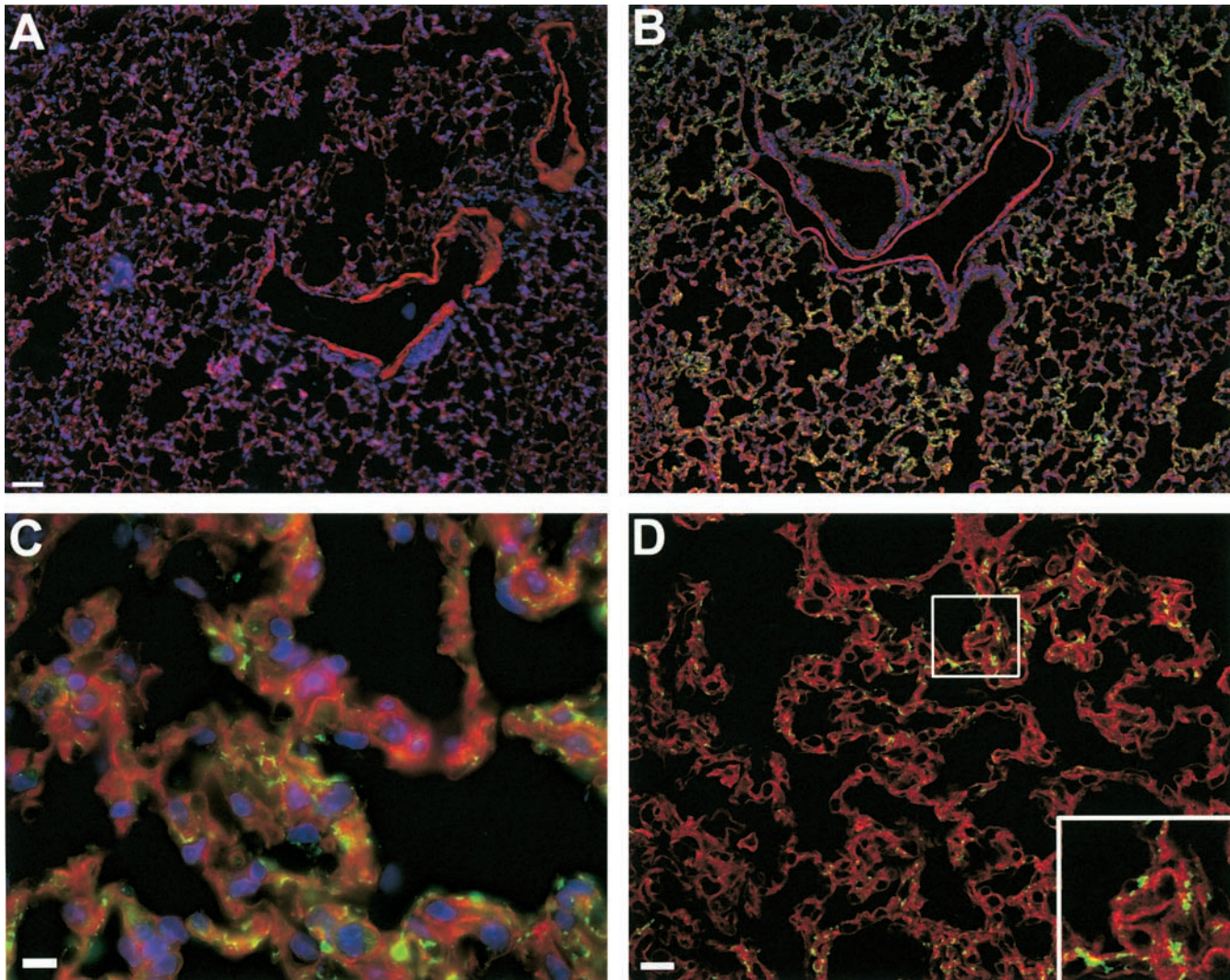


Figure 5. Tissue distribution of ODN 30 min following intravenous injection of LPD. BrdU-labeled ODN were formulated in LPD composed of DOTAP liposomes and injected into mice at a dose of 25 μ g/mouse. At 30 min following injection, mice were killed, lungs were collected, and cryosections were prepared. Detection of BrdU-labeled ODN in lungs was performed as described in MATERIALS AND METHODS. (A) Control lungs of mice receiving nonlabeled ODN. (B) Representative lung sections of mice receiving BrdU-labeled ODN. (C) Higher magnification of B. (D) Confocal microscopy images of lung sections of mice receiving BrdU-labeled ODN.

due to the unmethylated CpG motifs (13, 14). One possible approach to resolve this problem in antisense development is to choose ODN that lack a potent CpG motif(s). ODN 1–10 in Figure 7 are a few representative ODN that lack a CpG motif. ODN1668 is a CpG-containing ODN known to be potent in cytokine induction (15). The sequences of these ODN are shown in Table 1. In contrast to ODN1668 or plasmid DNA that was highly proinflammatory, ODN lacking a CpG motif were essentially inactive in TNF- α induction. These ODN were also inert with respect to the induction IL-1 and IFN- γ (data not shown).

In addition to induction of proinflammatory cytokines, systemic administration of cationic liposome/DNA complexes has been shown to induce other toxicities, including lymphopenia and elevations of serum transaminases (16, 17). A similar phenomenon was observed in this study fol-

lowing intravenous administration of LPD containing ODN 1668 (Figure 8). These hematologic toxicities also appear to be CpG-related, as intravenous administration of ODN 8 that lacks a CpG motif showed minimal levels of toxicities. DOTAP liposomes alone also did not induce significant toxicities at the dose examined (Figure 8).

Effect of LPD on Lung Permeability

Figure 9 shows the lung wet/dry weight ratios 12 h after intravenous injection of saline, liposome, or LPD containing either ODN 8 or ODN 1668. There is no significant difference among these groups although there appears to be a slight increase in the water content in the lung treated with ODN 1668. The lack of cytokine-induced increase in lung permeability in 1668-treated mice might be due to the transient nature of the cytokine response. Similar results were

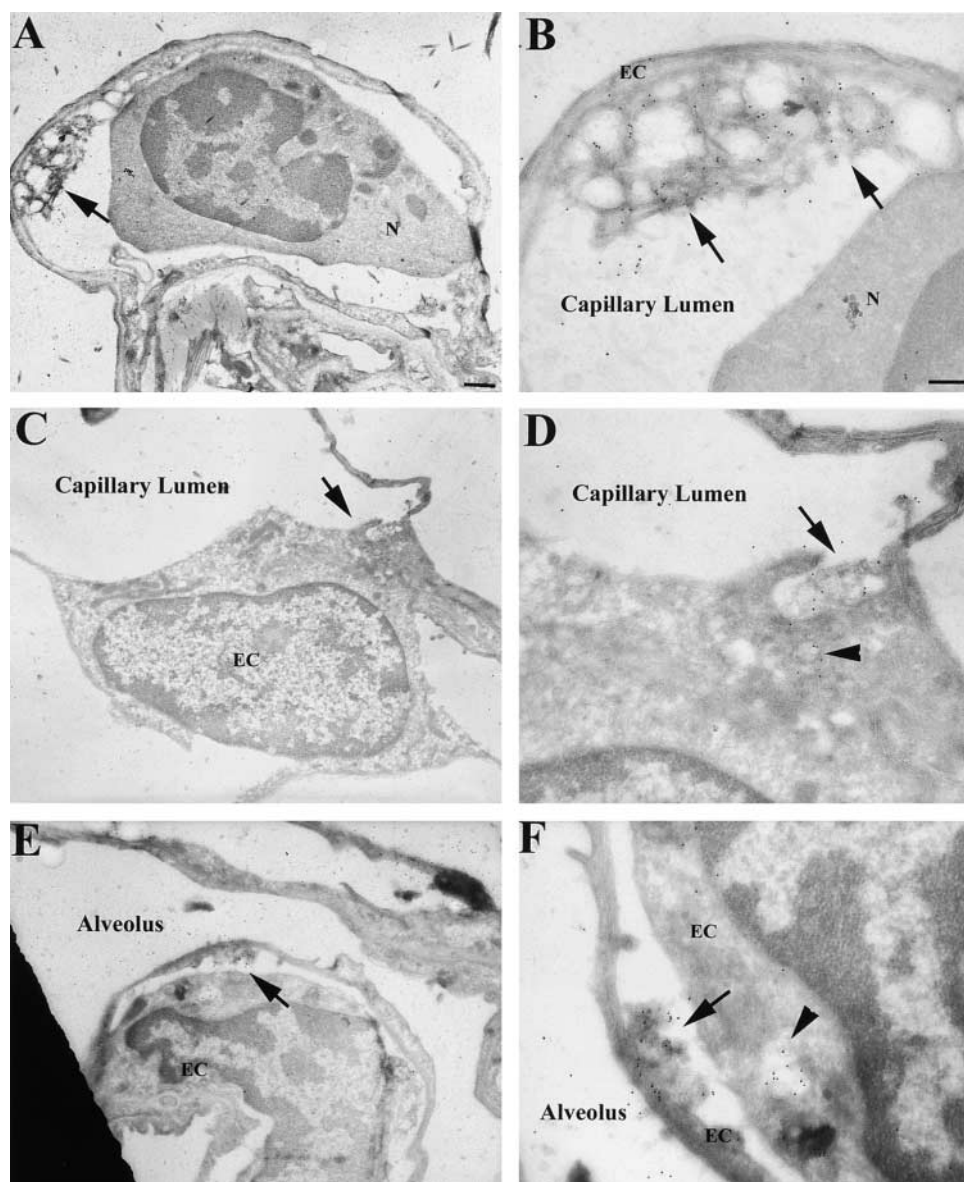


Figure 6. Immunogold localization of BrdU-labeled ODN in pulmonary endothelium after injection of BrdU-labeled ODN formulated in LPD. Tissue was processed and labeled as described in MATERIALS AND METHODS. (A) Five minutes after injection, gold label is shown together with lipidic vector adherent to surface of endothelium (arrow). Inside lumen of capillary is a neutrophil (N). (B) Higher magnification of A highlights gold labeled-ODN and lipid nature of complex (arrows) attached to EC luminal surface. (C) Thirty minutes after injection, ODN/lipid complex is observed being internalized (arrow) by the endothelium (EC). (D) Higher magnification of C details gold-labeled ODN and lipidic vector complex inside EC invagination. Gold-labeled ODN is also visible inside the EC cytoplasm, indicating cellular uptake (arrowhead). (E) Additional EC at 30 min following injection shows attachment of gold-labeled ODN/lipid complex to EC surfaces (arrow). (F) Higher magnification of E shows gold-labeled ODN attached to EC surfaces (arrow) as well as inside intracellular vesicles (arrowhead). Bar in A = 500 nm for A, C, and E; bar in B = 200 nm for B, D, and F.

observed when ^{125}I -labeled bovine serum albumin was used to evaluate the lung permeability (data not shown).

The Effect of ICAM-1 Antisense ODN on *E. coli* Endotoxin-Induced Expression of ICAM-1 mRNA

Endotoxin induced a significant increase in ICAM-1 mRNA in the lungs within 4 h of instillation (Figure 10). The increase in ICAM-1 mRNA was inhibited by pretreatment with LPD containing ICAM-1 antisense ODN (ISIS 3,082) at a dose of 2.5 mg/kg. Injection of free ODN at the same dose had no effect in decreasing the ICAM-1 mRNA (data not shown). Control ODN (ISIS 1,082) formulated in LPD had no effect either in downregulating ICAM-1 mRNA in mouse lungs. Repeating this experiment four times showed a range of 30–60% of downregulation of ICAM-1 mRNA by LPD/ISIS 3,082.

Discussion

We and others have previously shown that an efficient gene delivery to pulmonary endothelium can be achieved by intravenous administration of plasmid DNA formulated in cationic lipidic vectors (11, 12, 16, 18–20). In this study, we have demonstrated that cationic lipids can also mediate efficient delivery of ODN to pulmonary endothelium via the vascular route. The mechanism for cationic lipid-mediated delivery of nucleic acids via the vascular route is not clearly understood. Some studies have suggested that the proteoglycans on the cell surface play an important role in mediating the interaction with cationic lipid/DNA (ODN) complexes (21, 22). The efficient uptake of nucleic acids by lung might be due to the fact that lung is the first and largest capillary bed the cationic lipid/DNA (ODN) complexes will encounter following systemic administration.

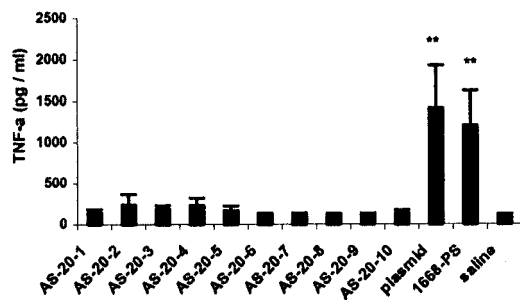


Figure 7. ODN lacking CpG motifs are non-proinflammatory following systemic administration. Groups of six mice received tail vein injection of ODN or plasmid DNA formulated in LPD at a dose of 25 μ g/mouse. Two hours after injection, blood was collected and the levels of TNF- α in serum were determined using a specific ELISA kit. The sequences of ODN examined were shown in Table 1. ** $P < 0.01$ (versus saline). $n = 6$.

However, the efficiency of DNA (ODN) delivery is also affected by the lipid composition. Previously, we have shown that LPD composed of DOTAP:cholesterol liposomes is the most efficient lipid vector for pulmonary gene transfer (10, 23). When LPD is evaluated for ODN delivery, LPD containing DOTAP alone appeared to be the most efficient formulation (Figure 2). One possible reason for this may be the structural differences between plasmid DNA and ODN. Plasmid DNA is a large molecule with a size of ~ 5 kb or greater. The mixing of plasmid DNA with

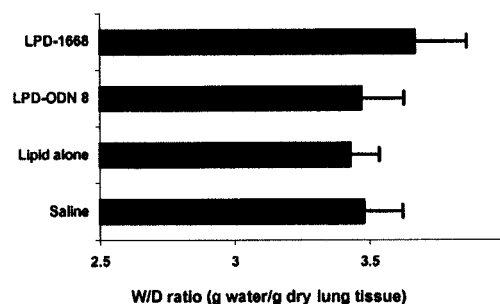


Figure 9. Lung wet/dry weight ratios following LPD treatment. Groups of six mice received tail vein injection of DOTAP liposome, or LPD containing either ODN 8 or ODN 1668. Twelve hours after injection, mice were killed and lungs were excised *en bloc* and dissected away from the heart and thymus. The lungs were immediately weighed and then placed in a desiccating oven at 65°C for 48 h, at which point dry weight was achieved. The ratio of wet/dry weight (g water/g lung dry weight) was used to quantify lung water content. $n = 6$.

protamine and cationic liposomes will lead to a substantial condensation of the plasmid (largely by protamine) along with lipid rearrangement (Li and coworkers, personal observation). In contrast, ODN are much smaller molecules and little, if any, condensation is expected. Indeed, preliminary studies demonstrated that cationic liposome/ODN complexes without protamine were as active as LPD in ODN delivery (data not shown). In the absence of condensation the affinity between ODN and cationic liposomes will be largely determined by the charge density on the liposome surface. Inclusion of any neutral lipid into the liposomes will decrease the surface charge density and therefore decrease their interaction with ODN. The decreased affinity of cationic liposomes for ODN will result in a rapid dissociation

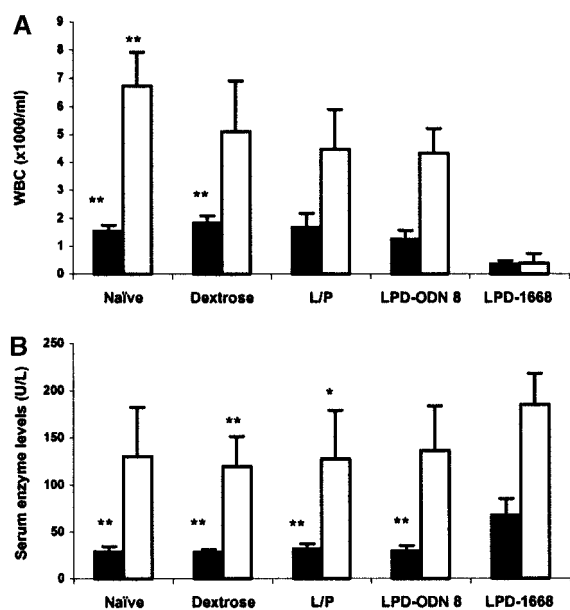


Figure 8. Whole blood and partial serum profiles after intravenous injection of dextrose, DOTAP liposome/protamine (L/P), or LPD containing either ODN 8 (LPD-ODN 8) or ODN 1668 (LPD-1668). Twenty-four hours later a complete blood count was performed; shown are partial white blood cell counts (neutrophils [filled bars] and lymphocytes [open bars]) (A). A small animal serology profile was also generated; shown are serum transaminases ALT (filled bars) and AST (open bars) (B). * $P < 0.05$ (versus LPD-1668); ** $P < 0.01$ (versus LPD-1668). $n = 6$.

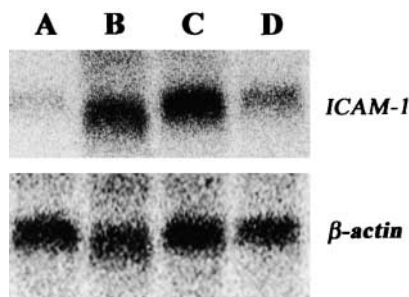


Figure 10. The effect of ICAM-1 antisense ODN on *E. coli* endotoxin-induced expression of ICAM-1 mRNA. Mice received an intravenous injection of ICAM-1 antisense ODN (ISIS 3082) or control ODN (ISIS 1082) formulated in LPD at a dose of 2.5 mg/kg. Thirty minutes after administering the ODN, endotoxin (80 μ l of 2.5 mg/ml) was instilled into the airways through the trachea. Control mice received saline. Four hours after the instillation, mice were killed and lungs were collected. Total RNA was isolated from lungs. Total RNA was separated on agarose/formaldehyde gels and transferred to nylon membranes and Northern blots were probed for ICAM-1 and β -actin mRNA using random primed labeled cDNA probes. A, saline control; B, LPS; C, ISIS 1082; D, ISIS 3082.

of ODN from LPD following exposure to serum (Figure 3) and, thus, inefficient pulmonary uptake. This is particularly the case with DOPE-containing LPD. DOPE is a lipid that contains a relatively small head group and two bulky acyl chains, which give the molecule an inverted cone shape. It is not favorable for the formation of a bilayer structure by itself under physiologic conditions. DOPE can form a bilayer with other lipids such as DOTAP; however, the resulting liposomes are relatively unstable and could undergo destabilization upon changes in pH, ion strength, temperature, etc. (24). Likewise, it is speculated that such unstable liposomes tend to form loose complexes with DNA (ODN) in which DNA (ODN) is poorly protected by cationic lipids. These loose structures may explain why DOPE-containing formulations have a rapid rate of disintegration following exposure to serum and is inefficient for delivery of both DNA and ODN via the systemic route.

Immunofluorescence staining of BrdU-labeled ODN suggested efficient accumulation of ODN in the alveolar capillary region and in small vessels (Figure 5). Transmission electron microscopy of immunogold localization of BrdU-labeled ODN (Figure 6) confirmed that pulmonary endothelial cells were indeed targeted by the vector. ODN were clearly visualized to be associated with or in the cytosol of the majority of endothelial cells examined. ODN were also found in other cells such as macrophages but at much lower frequencies. The study by Litzinger and coworkers using DC-chol:DOPE liposomes showed little uptake of ODN by pulmonary endothelial cells (25). This may be explained by the fact that complexes of DC-chol:DOPE liposomes with either plasmid DNA or ODN undergo a rapid process of disintegration following exposure to mouse serum (Li and colleagues, unpublished observation). This may result in poor interaction with pulmonary endothelium and thus inefficient uptake by endothelial cells.

One of the problems that is associated with cationic lipid-mediated delivery of nucleic acids is the induction of proinflammatory cytokines (13, 14, 17, 26, 27). We and others have recently demonstrated that intravenous administration of DNA formulated in cationic lipidic vectors can induce the production of a number of proinflammatory cytokines including IFN- γ , TNF- α , IL-12, and others (13, 17, 27). This cytokine response, which is DNA dose-dependent, is largely due to the unmethylated CpG motifs in the DNA (13, 14, 17, 26, 27). Indeed, a potent cytokine response was also noticed following systemic administration of complexes of cationic liposomes with an ODN containing a potent CpG motif (28). Although this cytokine response may prove to be beneficial for the treatment of tumors (28), it needs to be well controlled when ODN are used to treat nonmalignant pulmonary diseases or to characterize the function of novel genes. Recently several approaches have been proposed to decrease the CpG-mediated cytokine response such as the development of endothelium-specific vectors (29), the use of an immunosuppressant (27), and elimination of CpG motifs (14). Although the first two approaches may prove to be useful for DNA delivery, the last approach may only have limited success as not all of the CpG motifs are amenable to mutation. Mutation of the sequences that are critical for gene expression may result in

the inactivation of the expression plasmid. In contrast, all of these approaches are likely applicable to lipid-mediated ODN delivery. Because the optimal length of ODN is only 18–20 nucleotides and there are usually multiple targets in a given mRNA (30), it is likely that an antisense ODN will be identified that is highly active but does not contain a potent CpG motif. Such ODN will be much less inflammatory than either DNA or an ODN that contains a potent CpG motif. This was confirmed in this study (Figure 7). ODN 1–10 are a few ODN that lack a CpG motif. ODN1668 is a CpG-containing ODN known to be potent in cytokine induction. Indeed, ODN lacking a CpG motif were essentially inert in cytokine induction compared with plasmid DNA or a CpG-containing ODN. These ODN also induced minimal changes in blood counts and serology profile (Figure 8).

LPD induces minimal changes in lung permeability. Only a slight increase in water content was observed in lungs treated with ODN 1668 (Figure 9) despite the fact that it triggered a strong cytokine response (Figure 7). The lack of a cytokine-induced increase in lung permeability might be due to the transient nature of ODN-induced cytokine response. It remains to be determined whether ODN 1668 will induce a significant increase in lung permeability at higher doses.

To demonstrate the potential application of LPD in the treatment of pulmonary diseases, we investigated whether pretreatment of mice with LPD containing ICAM-1 antisense ODN (ISIS 3,082) can inhibit the LPS-induced ICAM-1 expression in mouse lungs. ISIS 3,082 is an ICAM-1-specific antisense ODN developed by ISIS Pharmaceuticals, Inc. (Carlsbad, CA) (31, 32). An early study by Kumasaka and coworkers showed that ISIS 3,082 alone could inhibit ICAM-1 expression in mouse lungs following LPS challenge (32). However, a high dose (30–100 mg/kg) was required to be effective (32). Our study showed that ISIS 3,082 formulated in LPD efficiently decreased the ICAM-1 mRNA at a dose as low as 2.5 mg/kg (Figure 10). Preliminary study showed that LPD-mediated delivery of ODN to the lung could also be achieved after LPS challenge (data not shown). We are currently evaluating the efficacy of post-treatment of LPD/ISIS 3,082 in downregulating the LPS-induced ICAM-1 expression.

Several delivery systems have been developed for ODN delivery, including peptide-ODN conjugates, antibody-polylysine conjugates and lipid-based vectors (33). One of the major advantages of lipidic systems is their large loading capacity. Thus, ODNs targeted to different genes could be codelivered to target cells. This may prove to be important in treatment of those diseases that involve upregulation of more than one gene.

Oligonucleotide technology has progressed very rapidly with the advancement in ODN synthesis chemistry and a better understanding of the mechanism of action of ODN. New generations of ODN have appeared that are more efficient in inhibiting the expression of a target gene and less toxic *in vivo* (5–7). These advancements, together with the development of more efficient delivery vehicles, will greatly expedite the development of ODN as a therapeutic agent as well as a research tool for gene function studies.

In summary, we have developed a novel lipidic vector composed of DOTAP liposomes, protamine, and ODN that is highly efficient in delivering ODN to pulmonary en-

dothelium via the vascular route. Furthermore, we have shown that this formulation is associated with minimal levels of proinflammatory cytokines and other hematologic toxicities when the ODN lack a potent unmethylated CpG motif. These results provide a basis for lipid-mediated delivery of ODN for the treatment of pulmonary diseases. They also suggest the utility of this approach as a research tool to characterize the function of genes in the pulmonary endothelium.

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